INTELLIGENT SYSTEMS / SEMINAR 1

# Reinforcement learning with the Astronaut

Training an astronaut to fly, dodge asteroids and collect oxygen using Q-learning

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**FLIGHT**
**100.0%**
Games completed without a collision

**+ ASTEROIDS**
**98.6%**
Games completed without a collision

**+ OXYGEN**
**96.8% / 5.32 tanks**
Games completed / average tanks per game

Five independent training runs; for asteroids and oxygen, each trained agent was tested in 100 unseen games.

## 01 How the agent learns

We begin with flight control, then add asteroids and oxygen tanks. Each stage builds on the Q-values learned in the previous one.

### WHAT GOES INTO THE Q-TABLE?

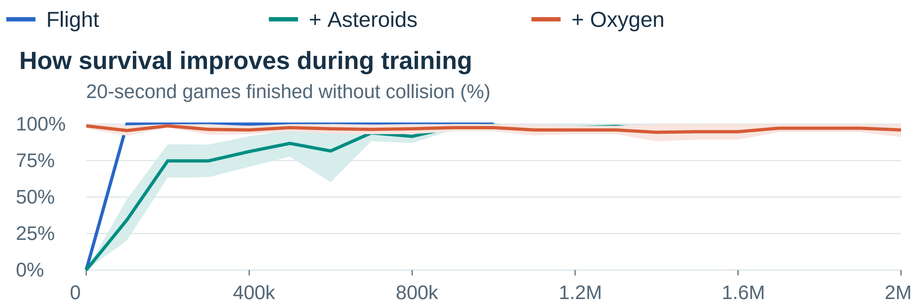
| | |
|---|---|
| **Flight**<br>Height and vertical speed | **72 states** |
| **+ Asteroids**<br>Flight + asteroid arrival time and height | **2,880 states** |
| **+ Oxygen**<br>Avoidance + oxygen position and availability | **103,680 states** |

For each state, the Q-table stores two values: one for coasting and one for thrusting. The astronaut tries both and updates the values based on what happens next.

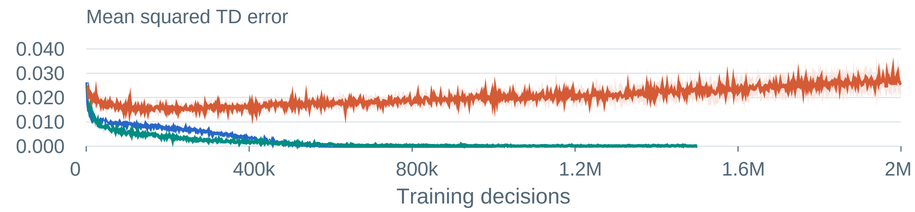
```
Q(s,a) <- Q(s,a) + alpha * error
```

The update uses the rewards received and the estimated value of the best next action. Each choice lasts up to six physics frames. A tank gives +1, a collision gives -1, and other steps give 0.

## 02 What changes during training




Each color shows a game stage. The x-axis counts action choices during training; the y-axis shows the percentage of 20-second games finished without a crash. At checkpoints, the agent plays without random exploration. Flight improves first; later stages build on earlier skills. Shading shows differences between agents.




Flight and asteroid prediction errors fall. Oxygen error remains higher and rises later, even though the agent collects tanks. We cannot claim loss convergence for oxygen.

## 03 How it plays after training

After training, the agents play using only their learned Q-values. 'Completed' means finishing all 20 seconds without a collision; 'Average game' is the mean survival time. The oxygen stage also reports tanks collected per game. Values are averages across five trained agents (+/- SD).

**Flight**
Completed: 100.0 +/- 0.0%
Average game: 20.00 seconds

**+ Asteroids**
Completed: 98.6 +/- 2.6%
Average game: 19.88 seconds

**+ Oxygen**
Completed: 96.8 +/- 3.1%
Tanks/game: 5.32 +/- 0.61

## 04 Conclusion

We trained the astronaut with tabular Q-learning, first to fly, then to avoid asteroids and collect oxygen. The hardest part was describing moving objects in a small enough state table without losing important detail. We are proud that the final agents finished 96.8% of test games and collected 5.32 tanks per game using the original rewards.

### TRAINING TIME: HOW LONG DID LEARNING TAKE?

